

REFERENCES: Chapter 2, and section 3.2 of KT

Complexity of bubble sort: $O(n^2)$
"worst case"

• WORST CASE ANALYSIS OF ALGORITHMS:

Q. How many graphs are possible on n vertices?

ans. $2^{\binom{n}{2}}$

	I_1	I_2	...	$I_{m'}$
instances (data)				
A_1	t_{11}	t_{12}		$t_{1m'}$
...				
A_i				
...				
A_m				

← running times

t_{ij} = max. of these run times denotes worst case for alg. A_i

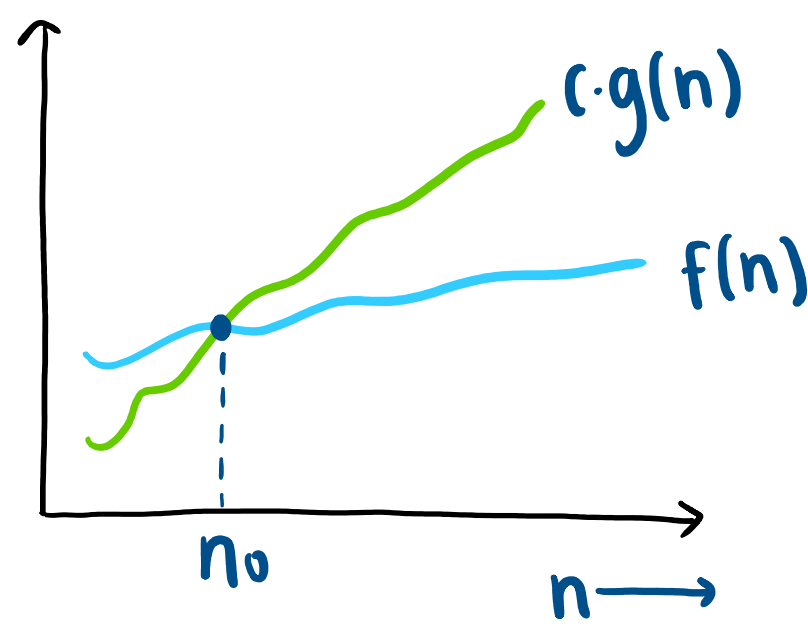
best algorithm $\Rightarrow$ min. of the max. values of algorithms

• ASYMPTOTICS:

* Big Oh Notation
= an analytical notation to hide some info.
= not really "worst case"

$O(g(n)) \Rightarrow f(n) \leq c \cdot g(n) \quad \forall n > n_0$
 $\Omega(g(n)) \Rightarrow f(n) \geq c \cdot g(n) \quad \forall n > n_0$
 $\Theta(g(n)) \Rightarrow c_1 \cdot g(n) \leq f(n) \leq c_2 \cdot g(n) \quad \forall n > n_0$

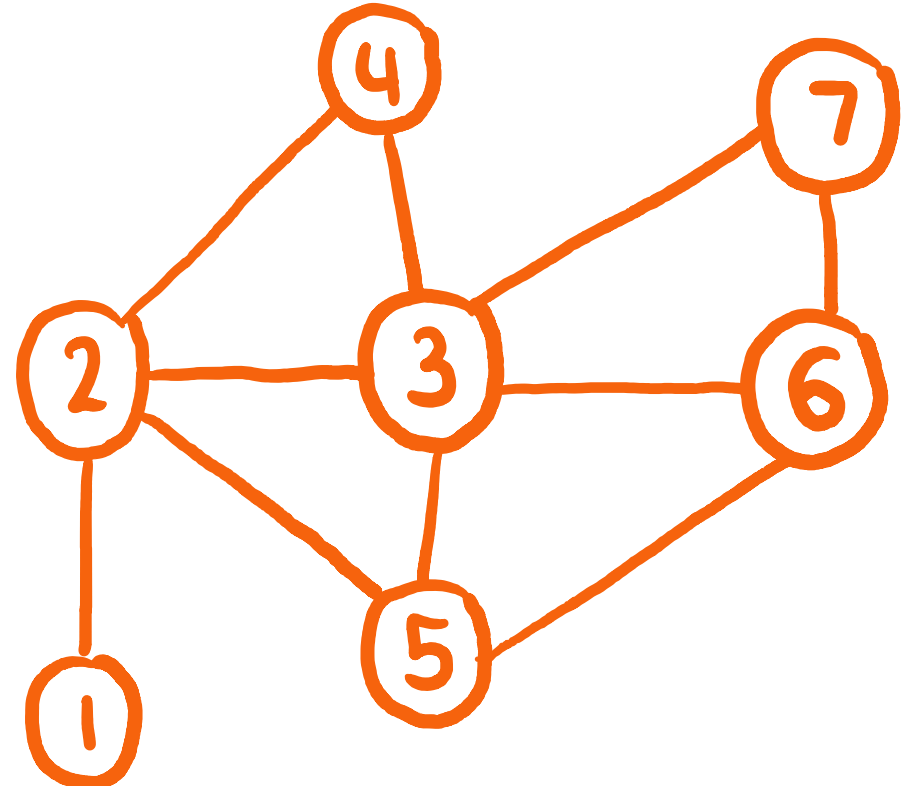
Let f and g be functions from $\mathbb{R}^+ \rightarrow \mathbb{R}^+$.
We say that $f(n) = O(g(n))$ if $\exists$ a constant c and n_0 large enough such that
 $\forall n \geq n_0, f(n) \leq c \cdot g(n)$



$f(n) \leq c \cdot g(n)$
 $\forall n > n_0$

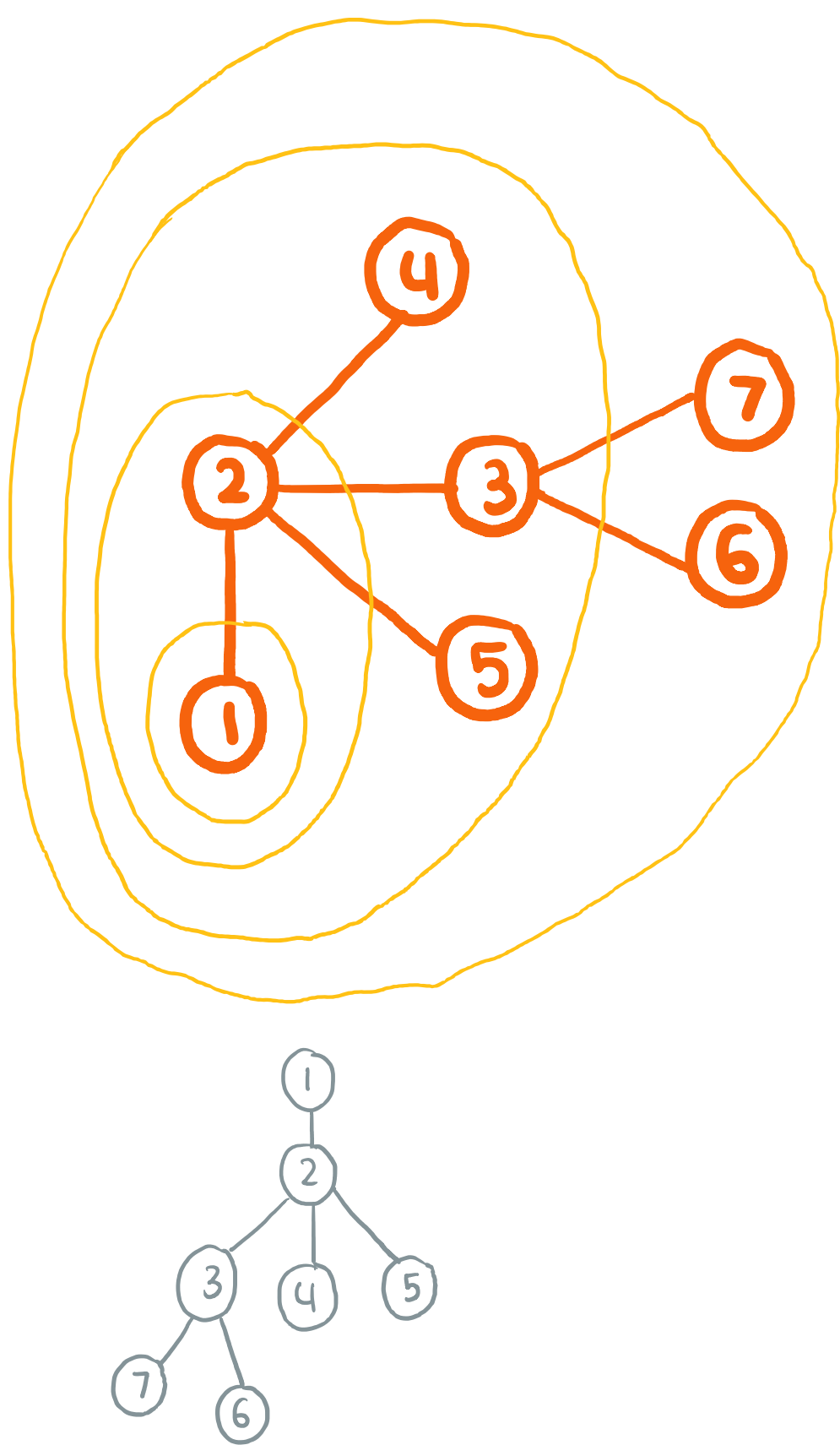
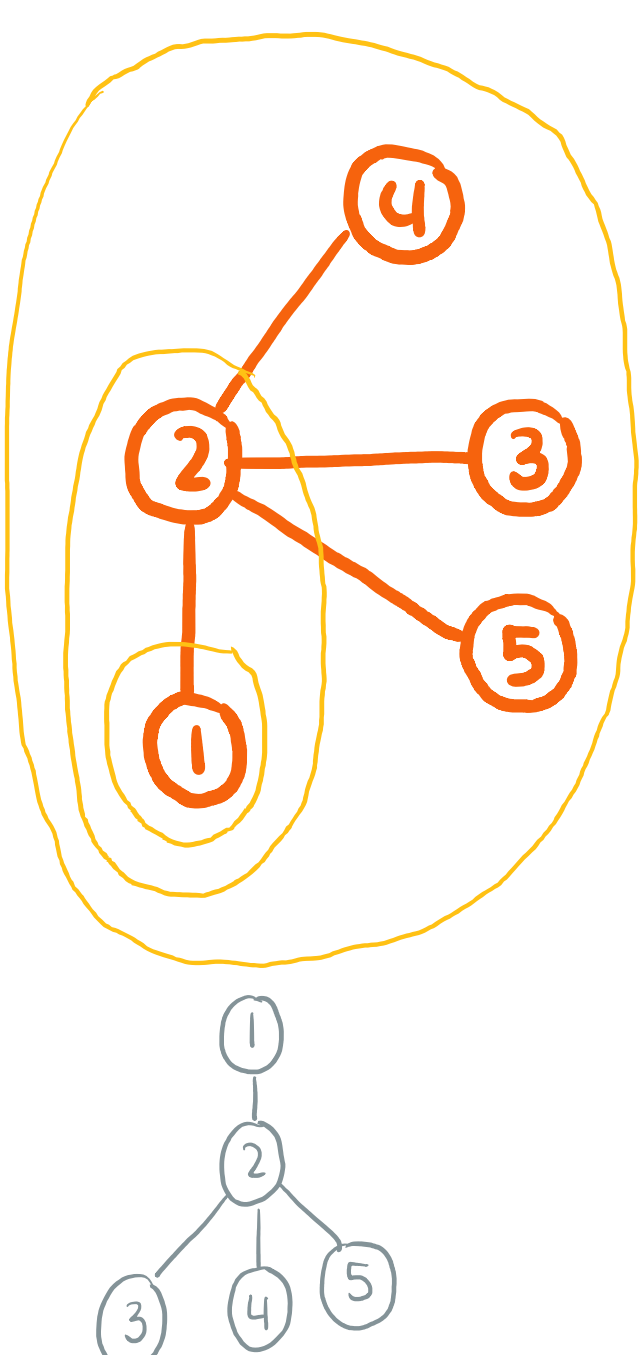
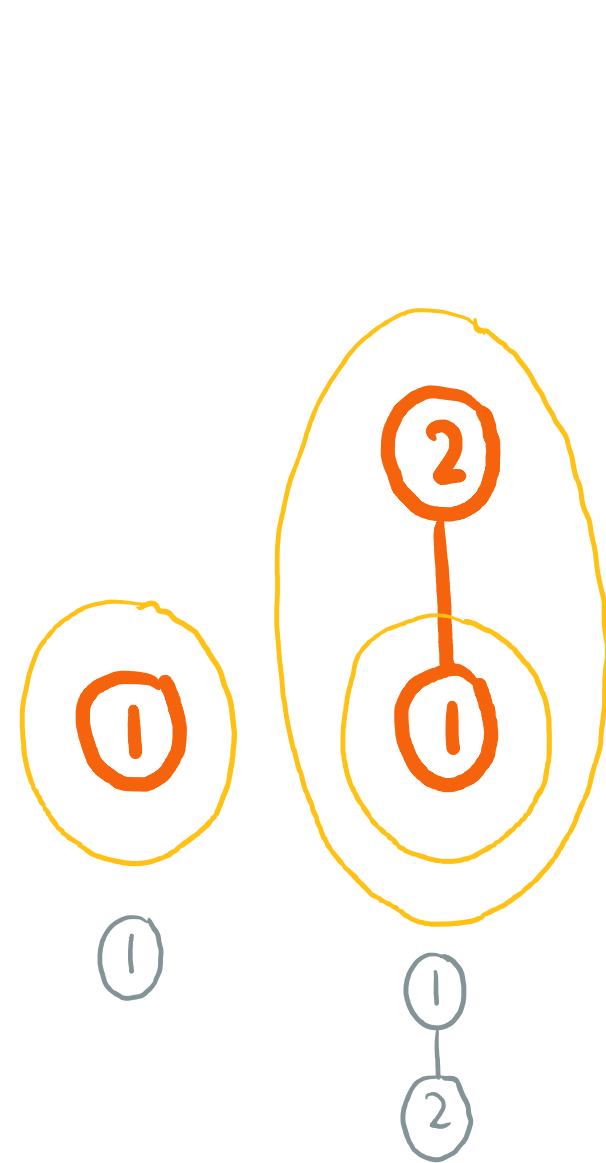
• BASIC GRAPH ALGORITHMS

Breadth First Search



$n \leftarrow$ # of vertices
 $m \leftarrow$ # of edges

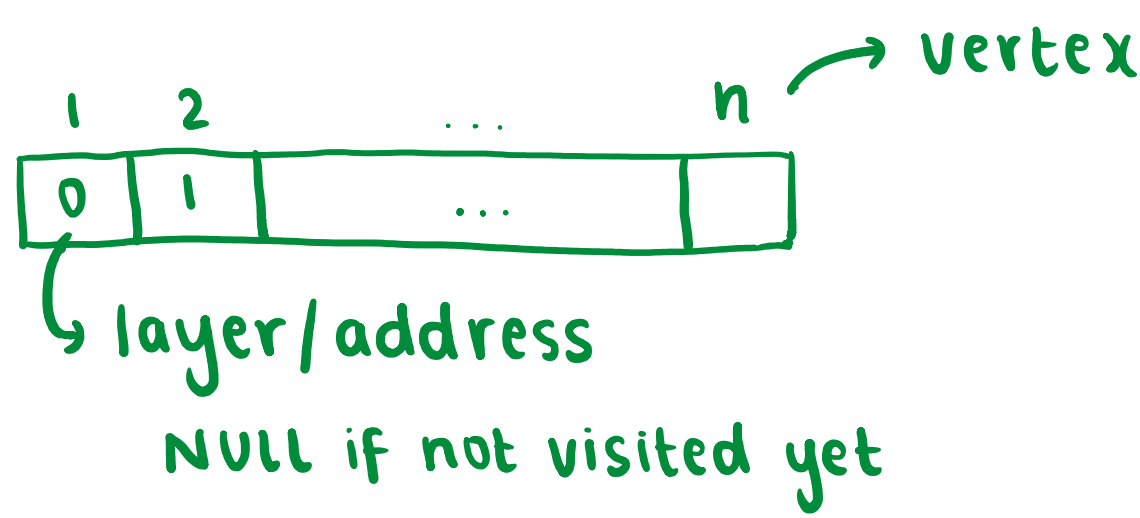
not every edge in the graph is represented.



Discovery Map / Discovery Tree

- Layered tree
 - Layer 0 contains the start node.
 - Layer 1 contains the neighbours of start node.
 - $\forall j \geq 2$, Layer L_j contains all vertices:
 - that are not already in $L_1, \dots, L_{j-1}$
 - and that have an edge to a vertex in L_{j-1} .

• BOOKKEEPING : explored / visited ?



For every vertex visited, we examine the neighbors.

$\sum d_v = 2|E| = 2m$

Running time
 $O(m+n)$

OBSERVATION : Layer numbers implicitly hold the information about the "shortest distance of a vertex" from the root.

Say this is false for the sake of contradiction.

THEOREM : All elements in Layer L_j have a shortest distance of j from the root.

Layer 0 = start node

BASE CASE: Contents of layer 1 are neighbors of the root node.
Say the theorem holds up to L_{j-1} .

Inductive step: Consider $v \in L_j$
 u is a predecessor of v such that $u \in L_{j-1}$
and $(u, v) \in E$
 $v \notin L_1 \cup \dots \cup L_{j-1}$

Let d be shortest distance of v from root.

CASE 1: $d < j$
not possible since $v \notin L_1 \cup \dots \cup L_{j-1}$

CASE 2: $d > j$
not possible since $v \in L_j$.

$\therefore d = j$